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CAPSTONE PRSENTATION

Introduction

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Title and Introduction

The goal of my project is to create a chatbot. Data was derived from Kaggle from a medical dataset that constitute different clinicians' assessment, the goal is to train via LLM and retrieve data using RAG and incorporate this on a python UI GRADIO (which is hosted on hugging face). Some Interesting facts about the Use of AI such as chatbots

- Advanced Models Hallucinate More: OpenAI's newer reasoning models (o3, o4-mini) hallucinated 33% to 48% of the time, more than double the rate of older, simpler models
- Heavy Users Encounter 3x More Hallucinations: Frequent AI users (over 6 hours/week) are nearly 3x more likely to encounter frequent hallucinations (34% vs. 12%) compared to light users
- "Longer" Prompts Are Riskier: Using very long prompts (e.g., entire documents or multi-source context) is 32x more likely to trigger frequent revisions compared to short, one-sentence prompts.
- Extremely High Hallucination Rates in Specific Fields: In specialized legal queries, state-of-the-art models have shown hallucination rates as high as 69% to 88%.
- Hallucinations are Almost Inevitable: standard AI training rewards "guessing" over fallback such as "I don't know," making confident, false answers mathematically likely.
- Confidence Is Not Accuracy: In one study, nearly 43% of AI-generated marketing content contained false or hallucinated information despite high confidence and high coherence
- 60% Hallucination Rate Doesn't Stop Adoption: Despite high hallucination users and consumers still find the output useful enough to continue using the tools
- 0% Data Leakage: Studies from March 2026, AI are good at securing data (do not leak confidential user data) but may invent facts to fill gaps in knowledge

Looking Ahead Project Goals outcome: My simple chatbot shows that RAG can affect hallucination based on the type of data (structured vs unstructured data) and the frequency of word during embedding because of semantic closeness and temperature. My chatbot shows RAG is far superior analyzing simple text file rather than data in a database

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Data and Overview

Source:

https://github.com/sebischair/medical-abstracts-tc-corpus?tab=readme-ov-file
https://www.kaggle.com/datasets/chaitanyakck/medical-text/code

Size:

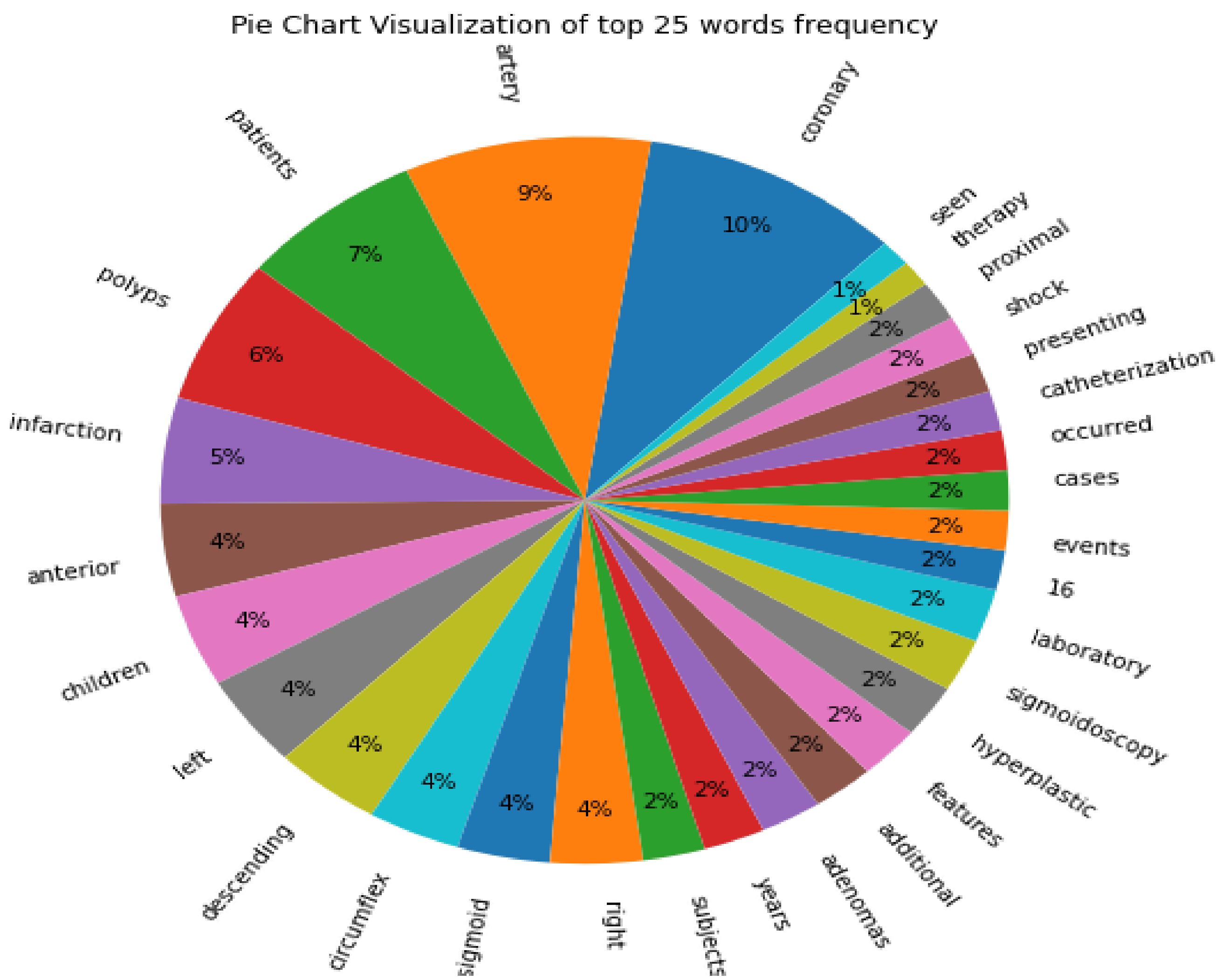
Text file has 17,811,684 Characters
2,612,370 words

Challenge:

- Running whole text dataset can become massive since chunking is involved.
- Main Point: Data is unstructured and can present potential problem if regular expression is not used
- Reason: This is used in simplifying dataset during embedding and chunking.

Goal: I analyzed a medical dataset that is unstructured to infer the capabilities of RAG by creating a chatbot and comparing generated response vs ground truth.

EDA Visualization



TFID Vectorizer showing the importance of the top 30 word to the first 3 papragraph while penalizing common words in determining semantic structure

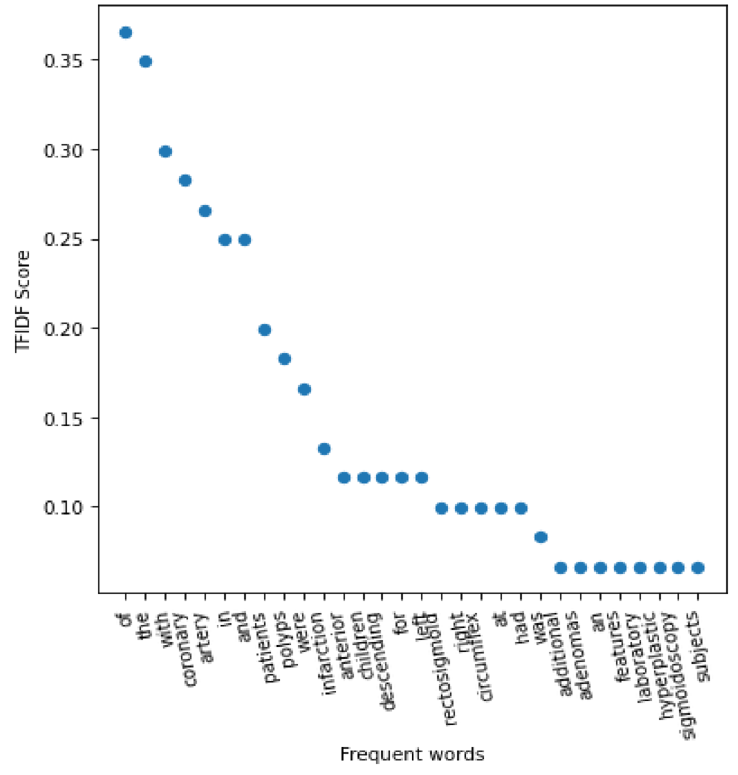


Table 1: First 7 frequent word as determined by the count vectorizer

Words	Count
Coronary	17
Artery	16
Patients	12
Polyps	11
Infarction	8
Anterior	7
Children	7

Table 2: First 7 word used with the top score used in determining semantic structure for RAG and their respective TFID-Score

Word	TFID-Score
of	0.365652
the	0.349032
with	0.299170
coronary	0.282550
artery	0.265929
in	0.249308
patients	0.199447

Problem Statement and Motivation

Context

To create a chatbot and implement NLP algorithm such as RAG (Retrieval Augmented Generation). This skills applies core concept of NLP discussed in previous project that analyze embedding via word2vec and vectorization and manual chunking using n-gram

Why this matters:

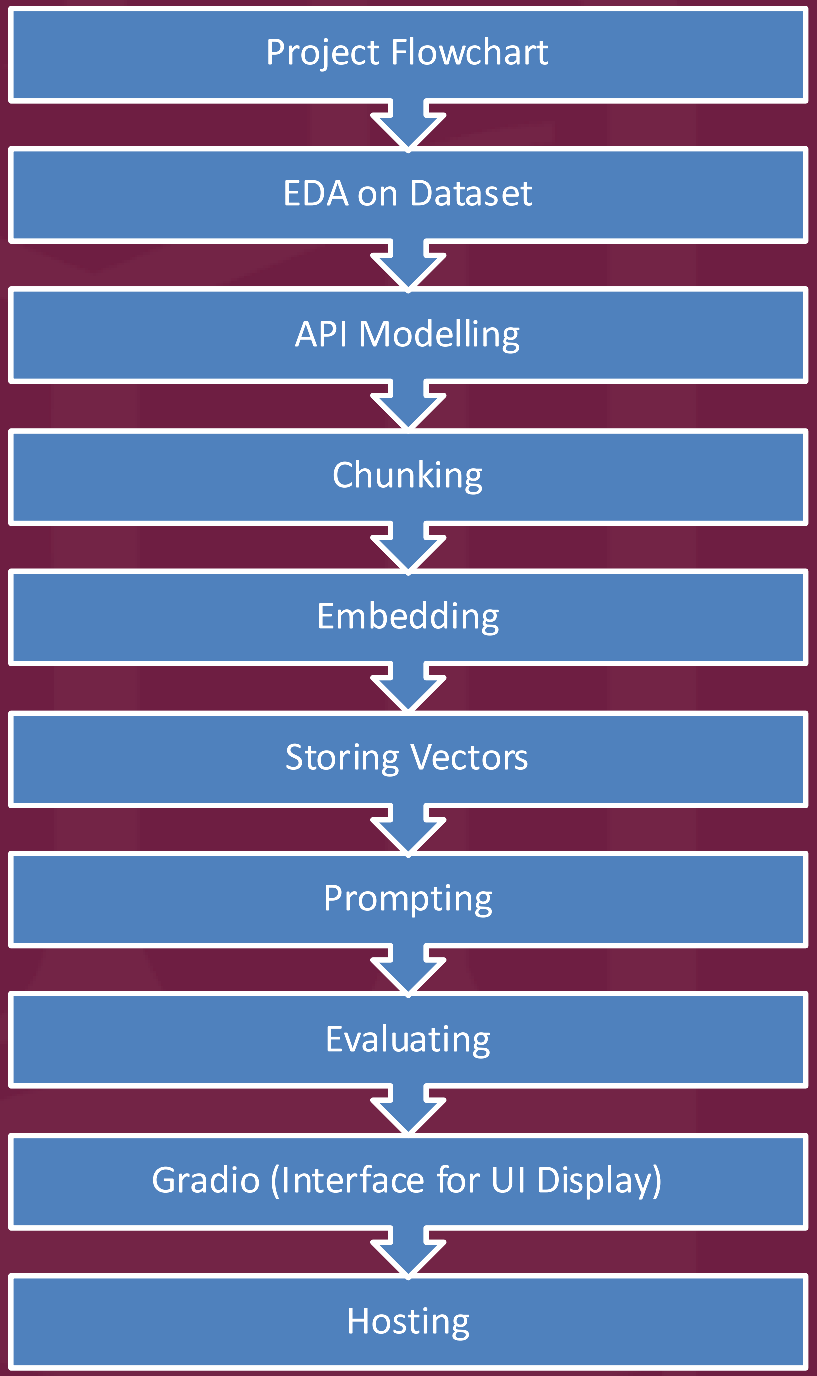
- AI generates revenue for companies in all sectors by cutting time, saving cost and improving efficiency.
- AI role is not to replace the need for human interaction but to assist in optimizing decision making as data become more vast and too much to hand.
- CNBC OpenAI expect to generate 12 billion by end of 2025 in total revenue, Anthropic generates about 3 billion in annualized revenue.

Citation: (https://www.cnbc.com/2025/05/30/anthropic-hits-3-billion-in-annualized-revenue-on-business-demand-for-ai.html)

The following objectives will be discussed

- Create a crude text chatbot that is hosted on hugging face and test its functionality, screen shot will be added to display its result
- Run evaluation metrics
- Determine the importance in RAG in text chatbots, improvement and limitation

Methodology and Approach



Project flowchart : Perform EDA → API model → Chunking → Embedding → Storing Vectors→ Prompting → Evaluating → Gradio → Hosting

Reproducibility:

- **GitHub: Code can reproduce by changing dataset and updating model, packages and import depending on type of coding software.**
- **https://github.com/isaacadebayo/CodingProjects/blob/main/gradio_rag_capstone_.py**
- **Other evaluation metrics can be used to determine hallucination of model, human metrics can also be incorporated to see how often the correct response is cited, likewise a code prompt can be added to show where chatbot got their exact information from for their abstract summarization which can be used to verify authenticity.**

RAG Pipeline

- ❑ EDA – After reading the dataset, Word frequency counter and TFID score will be used to analyze important word in the text that is involved in RAG
- ❑ API Model – Using API model, a key will be generated and the appropriate GPT model will be used from a varieties of model to train using large language model (LLM)
- ❑ Chunking – Chunking will be done using FAISS with 250 chunk size and 10 chunk overlap. This will be a recursive character text splitter.
- ❑ Embedding – Convert text to numerical variable using vector embedding derived from openAI model ("text-embedding-3-large")
- ❑ FAISS: Saves vectors into FAISS, allows you to find the most "similar" code instantly when a user asks a question.
- ❑ Prompt: Prompt with 4 different questions.
- ❑ Gradio: Python interface to visualize prompt, interface can be customized
- ❑ Evaluate: Evaluate bot generated response with ground truth

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Evaluation Metrics

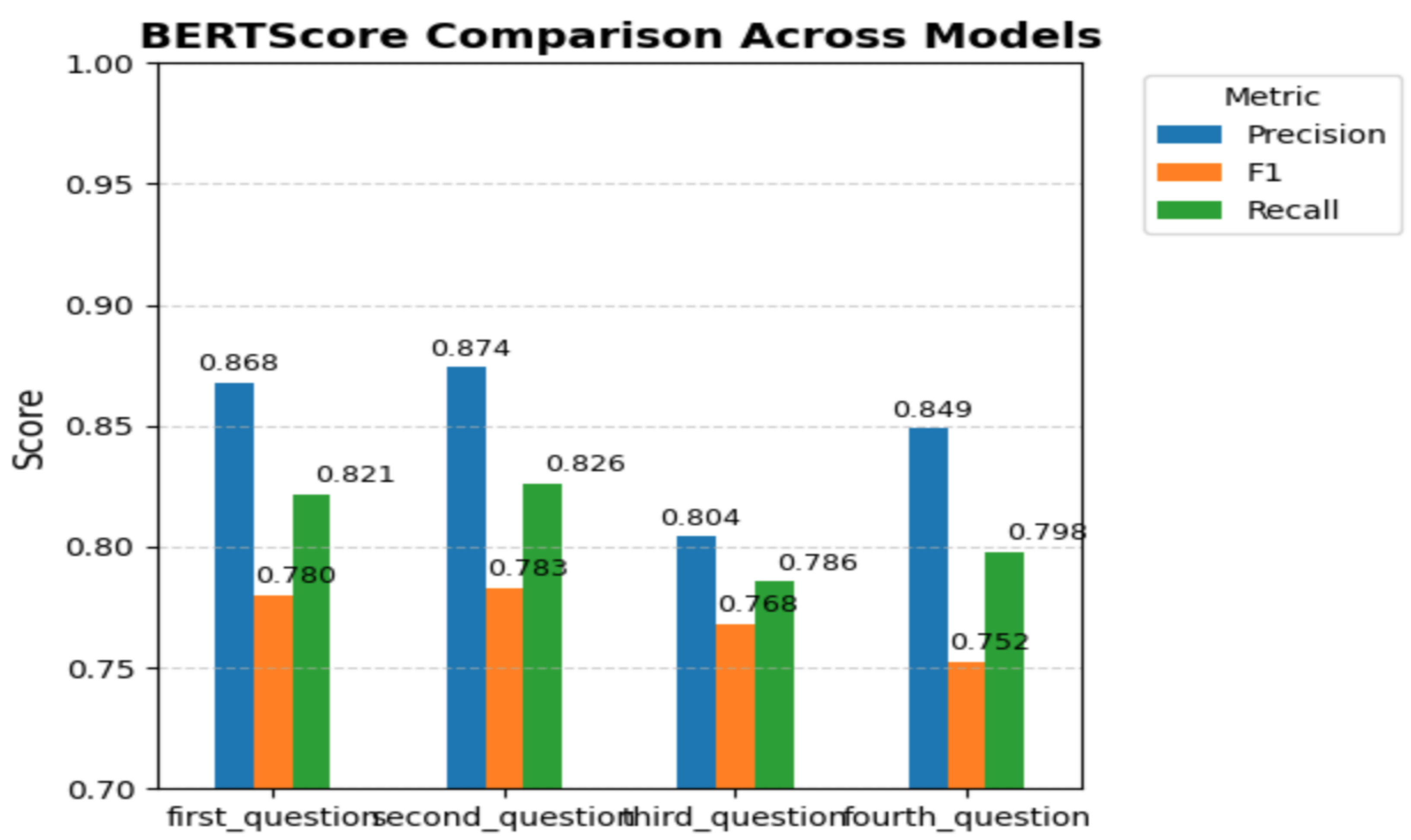
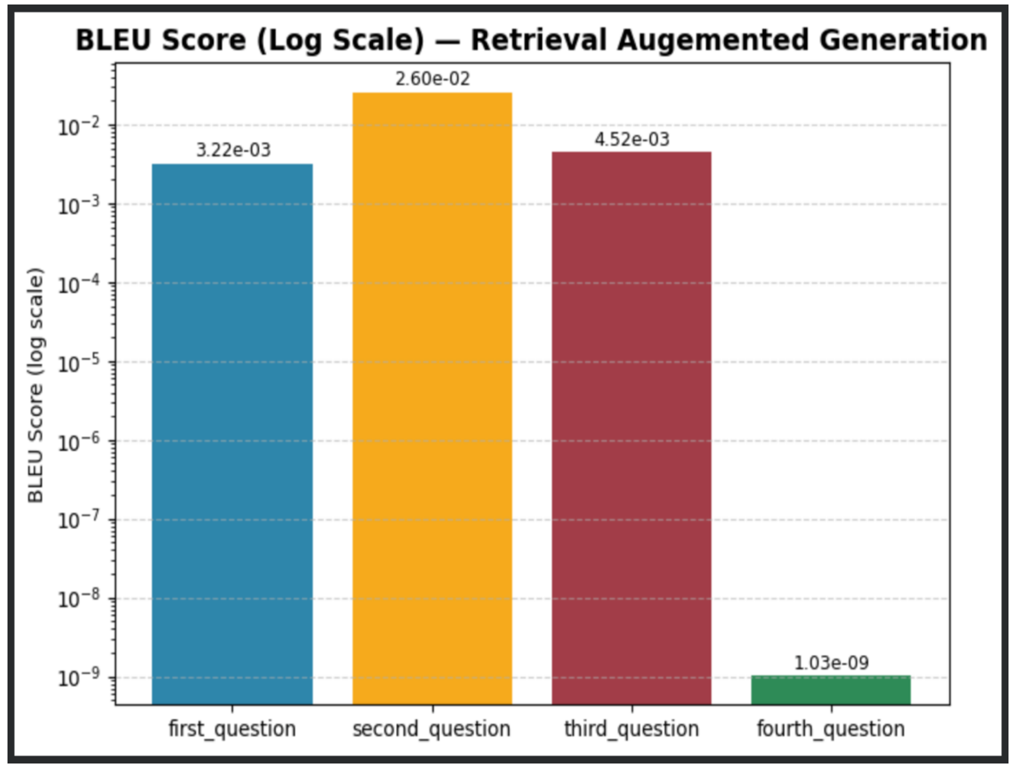
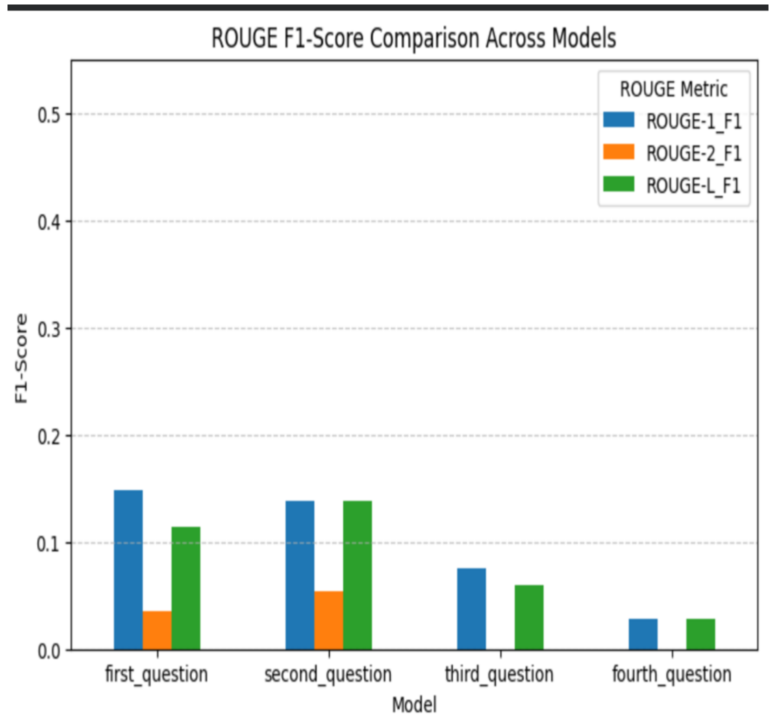


Table 3: Various evaluation metrics, their module or package name and their respective score for prompt .

Evaluation Metrics	Model/Package	Score
Fallback Rate	PyTorch	50.00%
Confidence Score	PyTorch	99.70%
Coherence Score	Deepval, Geval, LLMTestCase, LLMTestCaseParams	0.8808067206619494
Perplexity	gpt_2	146.38





Key Results

Table 4: BERT Score Precision, Recall and F1-Score for all four prompted question using the Roberta-large model

Model Name	BERT Evaluation Metrics	First question Score	Second question Score	Third question Score	Fourth Question Score
Roberta-large	Precision	0.8677	0.8740	0.8040	0.8488
	F1	0.7799	0.7827	0.7679	0.7523
	Recall	0.8214	0.8259	0.7855	0.7976

Evaluation metrics used

- ROUGE: Rely on n-gram matching, F1-scores were all low (around 0.1, 0.2) and were excluded from analysis.
- BLEU: Bad for abstractive summarization because algorithm tris to match the exact word leading to a low score.
- BERT Score: Take into consideration word context.
- Coherence: Metrics used to measure the clarity and flow of sentence structure in generated chatbot output response.
- Perplexity: The metrics used to measure the ability to generate novel words often tied with temperature.
- Fallback rate: The number of times the fallback statement ("I don't have enough information to answer that was used").

KEY Report

- Point: BERT scored high to assess the quality of test and semantic closeness retrieved by RAG
- Reason: This is useful in understanding not just retrieval but word context and generation.
- Point restated: BERT scored outperformed other evaluation metrics in evaluating similarity between a chatbot generated response and the ground truth.

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Discussion and Interpretation

```
[59]
%%time
first_question = ask_question("What were the ages of the patient in the Catheterization laboratory events and hospital outcome with direct angiopla
print(first_question)

• The patients in the study ranged in age from 31 to 85 years for the Catheterization laboratory events and hospital outcome with direct angioplasty
• Successful angioplasty was achieved in these patients, with residual stenosis less than 50% and no major complications.
CPU times: user 109 ms, sys: 132 µs, total: 109 ms
Wall time: 1.97 s

[60]
%%time
second_question = ask_question("What was the reason for conducting the Catheterization laboratory events?")
print(second_question )

• To assess the safety of direct infarct angioplasty without antecedent thrombolytic therapy
• To demonstrate anatomy and physiology in previously problematic conditions
• To evaluate symptoms and postmyocardial infarction testing in male patients
CPU times: user 98.7 ms, sys: 1.3 ms, total: 100 ms
Wall time: 1.6 s

[61]
%%time
third_question = ask_question("What was the culprit of infection in renal abscees in children for both male and female. How many cases were analyze
print(third_question )

* The culprit of infection in renal abscesses in children were:
   - Staphylococcus aureus in boys
   - Escherichia coli in girls
* A total of 26 cases were analyzed for clinical features in renal abscesses in children.
CPU times: user 107 ms, sys: 327 µs, total: 107 ms
Wall time: 1.62 s
```

Result Insights

- ❑ Each question prompt had different precision, recall and F1-Score. This is expected because BERT score is based on semantic closeness which is contextual and depends on the information processed. The second prompt question scored high on all metrics as seen in the result question

- ❑ A coherence score of the second question was found to be 0.897 against a threshold of 0.9 to check sentence structure and presence of contradictory information

- ❑ A perplexity score of 146 was found using a gpt-2 large model

- ❑ Result indicate chatbot performs well analyzing text data and generating response.

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Comparison with bot generated response and original data

infection. Antibiotics alone produced a cure in 10 children (38%), but 16 children (62%) required a surgical procedure.

2 Hyperplastic polyps seen at sigmoidoscopy are markers for additional adenomas seen at colonoscopy. Asymptomatic individuals undergoing screening flexible sigmoidoscopy were prospectively studied. Polyps were found in 185 subjects. The endoscopist recorded an opinion on the polyps' histology based on endoscopic appearance. No polyps were removed at sigmoidoscopy. All subjects with rectosigmoid polyps then underwent colonoscopy and polypectomy. Of them, 99 subjects (54%) had at least one rectosigmoid adenoma, 69 (37%) had only hyperplastic polyps, and 17 (9%) had other findings. The endoscopists' opinion of the histopathology of polyps at sigmoidoscopy was correct for 61% of the lesions. Of subjects with adenomatous rectosigmoid polyps, 29% had additional adenomas at more proximal sites. Proximal adenomas were found in 28% of patients with hyperplastic rectosigmoid polyps. Patients with rectosigmoid hyperplastic polyps had the same risk for additional proximal adenomas as patients with rectosigmoid adenomatous polyps.

4 Catheterization laboratory events and hospital outcome with direct angioplasty for acute myocardial infarction To assess the safety of direct infarct angioplasty without antecedent thrombolytic therapy, catheterization laboratory and hospital events were assessed in consecutively treated patients with infarctions involving the left anterior descending (n = 100 patients), right (n = 100), and circumflex (n = 50) coronary arteries. The groups of patients were similar for age (left anterior descending coronary artery, 59 years; right coronary artery, 58 years; circumflex coronary artery, 62 years), patients with multivessel disease (left anterior descending coronary artery, 55%; right coronary artery, 55%; circumflex coronary artery, 64%), and patients with initial grade 0/1 antegrade flow (left anterior descending coronary artery, 79%; right coronary artery, 84%; circumflex coronary artery, 90%). Cardiogenic shock was present in eight patients with infarction of the left anterior descending coronary artery, four with infarction of the right coronary artery, and four with infarction of the circumflex coronary artery. Major catheterization laboratory events (cardioversion, cardiopulmonary resuscitation, dopamine or intra-aortic balloon pump support for hypotension, and urgent surgery) occurred in 10 patients with infarction of the left anterior descending coronary artery, eight with infarction of the right coronary artery, and four with infarction of the circumflex coronary artery (16 of 16 shock and six of 234 nonshock patients, p less than 0.001). There was one in-laboratory death (shock patient with infarction of the left anterior descending coronary artery).

5 Renal abscess in children. Three cases of renal abscesses in children are described to illustrate the variable presenting features. An additional 23 pediatric cases, reported over the past ten years, were reviewed for clinical features and therapy. Fever, loin pain, and leukocytosis were common presenting features, but less than half of all abscesses were associated with either an abnormal urinalysis or a positive urine culture. The presenting features were sometimes confused with appendicitis, peritonitis, or a Wilms tumor. An organism was identified in 17 cases--Escherichia coli in 9 children and Staphylococcus aureus in 8 children. The majority of E. coli infections occurred in girls and the majority of S. aureus infections occurred in boys. Reflux was documented in 5 patients, and 2 children had a possible extrarenal source of infection. Antibiotics alone produced a cure in 10 children (38%), but 16 children (62%) required a surgical procedure.

Zoning in on RAG and Hallucination

```
[65]
%%time
fourth_question = ask_question("Polyps were found in how many subjects in the hyperplastic polyps?")
print(fourth_question)

• Polyps were found in 108 subjects with hyperplastic polyps.
CPU times: user 109 ms, sys: 2.49 ms, total: 112 ms
Wall time: 1.51 s
```

```
[64]
%%time
# Different Prompting
fifth_question = ask_question("In the Hyperplastic polyps seen at sigmoidoscopy are markers for additional adenomas. . How many people had polyps")
print(fifth_question)

- Polyps were found in 185 subjects.
CPU times: user 105 ms, sys: 110 µs, total: 105 ms
Wall time: 1.05 s
```

- **Embedding model: During the similarity search, RAG picks up on semantic closeness (similar words or phrases) even if those terms are being used in entirely different or mis-associated contexts across different paragraphs.**

- **Semantic Search Nuances: Embedding models are designed to understand the meaning of words and phrases. However, meaning is highly contextual.**

- **The 'Mis-association' Problem: When a LLM algorithm receives semantically similar but contextually mis-associated chunks, it tries to synthesize an answer from all of them. This can lead to:**
  - **Conflicting Information: The LLM might inadvertently combine information that should be kept separate.**
  - **Loss of Specificity: The generated answer might become overly generalized or ambiguous, losing the precise context of each retrieved piece of information.**
  - **Subtle Hallucinations: The LLM might infer connections or relationships between these mis-associated pieces of information that aren't truly present in your dataset, leading to subtle forms of hallucination**

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Future Work

Current dilemma: Balancing novelty creativity with generated response and factual-information

Since, RAG is prone to hallucination or misinterpretation when working with unstructured databases some common recommendation and active research are listed

Potential Cause of Hallucination

- Textualization challenge: RAG systems fundamentally operate on text. Database are structured. RAG needs to convert this structured data into a textual format which can lose context and introduce ambiguity for LLM
- Schema Misinterpretation: Database require understanding relationships between tables or specific column. If the RAG system attempts to translate a natural language query into a database query (e.g., SQL), hallucination can occur.
- Misassociation on Values: Similar values that appear in different, unrelated contexts across various tables or rows, the textualized chunks might become very similar. The RAG retrieval might then incorrectly pull together these mis-associated pieces of information, causing hallucination
- Granularity Issues: Incorrect granularity (level of detail) can either provide too little context or too much noisy, irrelevant context, both of which can increase hallucination risk.

Contrast with Unstructured Text Files

- Native Format: For unstructured text files, the data is already in the format RAG expects. The main challenge is effective chunking and ensuring each chunk is a self-contained unit of meaning.
- Richer Context: Text files often contain more natural language narratives that implicitly define relationships and context, making it easier for the LLM to understand.

Limitation and Ethics

DATASET for this project is Publicly available, de-identified datasets are stripped of personal identifiers (names, SSNs, etc.) to remove individual identification risk, often exempting them from Institutional Review Board (IRB) review under 45 CFR 46. These datasets are deemed safe for research, public use, and scholarly sharing."

LINK: https://ovpr.uconn.edu/services/rics/irb/researcher-guide/secondary-analysis-of-data-sets/

Ethical, Privacy and Bias concerns.

Privacy

- The collection of data poses risk. Communication between a chatbot and user involves interaction between a user and the LLM.
- Chat logs could contain sensitive or personal queries such as feeling, mood or emotion. AI needs to analyze when to report or flag safety concerns

Risk of Hallucination can be detrimental

- Taking information as face value can lead to negative outcome, over-reliant of AI hallucinated as been linked to AI induced psychosis.

Ethics implementation

- Ethical rules will be incorporated in the training program to prevent the chatbot from making unfounded assumptions when prompted with queries requiring professional expertise.
- A fall back "If unsure, say "I don't have enough information to answer that" or "I am not a licensed trained professional, but I can direct you to one is recommended.

Strength and Honest Reflection

- Huge tremendous benefit of AI to improve efficiency. AI excels in summarization of dense material in different fields.
- AI chatbot rarely struggle with established facts.
- AI chatbots are conduit for improvement meaning they provide information that meant for further research.
- AI become problematic when it comes to creative thinking. AI can generate novel response which can spiral into a list of hallucinated work.
- My final analysis: AI is a useful breakthrough in the computational field like how penicillin was in the 20th century medical field. Over-reliant and overuse of such tools can become detrimental with times as knowledge evolve.

Conclusion and What's next for my LLM Chatbot

CAPSTONE RECAP

- The main takeaway of this project was to build a chatbot and investigate the use of RAG and LLM in retrieving and generating new response output.
- The project identify the retrieval component as one of the major source of hallucination. Some recommendation includes advanced chunking and hybrid search

Recommendations

- AI companies should educate users on prompting techniques when interacting with AI with the the objective to reduce hallucination.
- Self initiatives: Free coding resources such as Sololearn are available that offer a crude introduction on AI infrastructures and framework (prompt engineering).

Future Work and improvement on my chatbot capstone should include :

- Advanced Chunking: Semantic chunking, hierarchical chunking, or document-aware chunking are recommended
- Hybrid Search: Combine semantic search with keyword search This can help ground the retrieval more precisely.
- Query Expansion/Rewriting: Use an LLM to expand or rewrite the user's query to make it more precise and less ambiguous, leading to more targeted retrieval.

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Workflow Citation

Libraries Citation
- Gradio
- langchain langchain-openai faiss-cpu pypdf
- langchain langchain-community
- !langchain langchain-huggingface
- bs4
- langchain
- langchain-core
- openai
- sentence-transformers
- matplotlib.pyplot as plt
- seaborn
- numpy
- sklearn
- sklearn.feature_extraction.text import CountVectorizer
- sklearn.feature_extraction.text import TfidfVectorizer
- sklearn.feature_extraction import text
- pandas as pd

Workflow Citation
- OpenAI platform
- Python (Google Colab)
- Hugging Face
- FAISS
- Gemini

Evaluation Metrics Libraries
- import re # Regular expressions
- import nltk # Tokenisation
- from transformers import pipeline
- rouge-score
- ! sacrebleu
- bert-score
- transformers torch
- ragas
- deepeval
- transformers torch

Question and Reference

Thanks for your attention and time

The floor is open to questions

DEMO

https://huggingface.co/spaces/IsaacAdebayo/chatbot?logs=build

https://colab.research.google.com/drive/1TfLbJaWjg4U63U9_9_czeSD9yk1OjlGS#scrollTo=E9IARCIkga0L